



UNIVERSITÀ DI PARMA

# Laboratori & code::blocks

*Per me si va ne la città dolente,  
per me si va ne l'eterno dolore,  
per me si va tra la perduta gente*

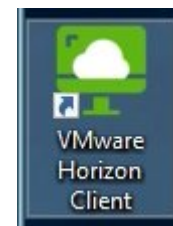
Dante Alighieri, Inferno Canto III

- Utilizzo laboratori
  - Accesso
  - Accesso remoto
- Code::blocks
  - Cosa è
  - Utilizzo

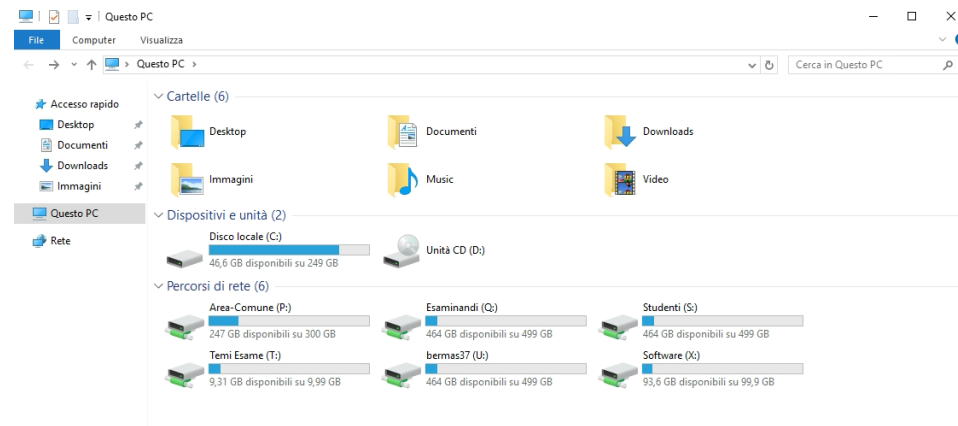
SUMMARY



- Credenziali
  - nome.cognomeXYZ@studenti.unipr.it
  - Relativa password
  - Dominio UNIPR (ma è automatico)
  - In tutti i laboratori
- Dopo accesso, se macchina fisica, occorre usare VMware Horizon Client
- Per esame → credenziali generiche (isdXesaY/“noncopiare”)



- Share di rete
  - U: vostra home
    - Salvate i vostri dati sempre qui
    - C: dipende da macchina
  - T: area docenti/esami
    - Materiale docenti
    - Temi esame ecc.

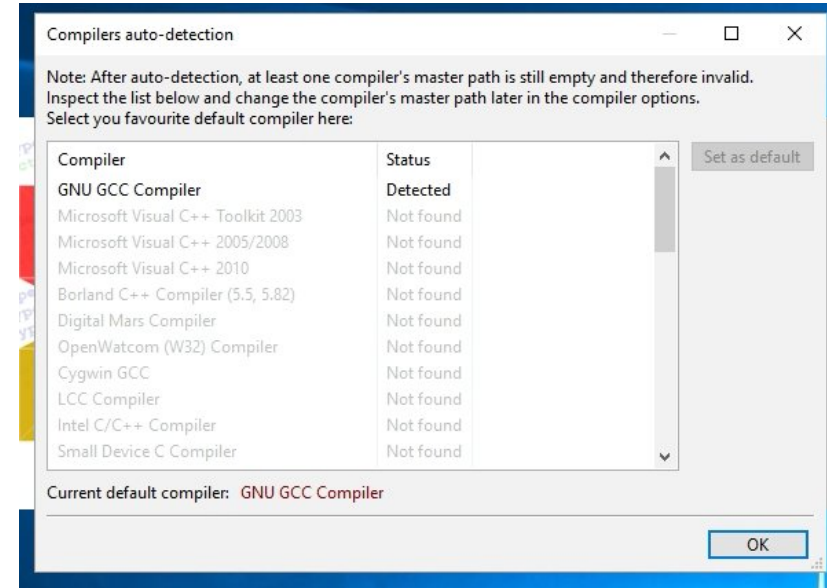


- Possibile accesso da remoto
  - VMware Horizon Client
  - Istruzioni e download in:
  - [https://wiki.asi.unipr.it/dokuwiki/doku.php?id=guide\\_utente:accesso\\_remoto\\_laboratori](https://wiki.asi.unipr.it/dokuwiki/doku.php?id=guide_utente:accesso_remoto_laboratori)
- Sempre accedere a vdi-gw.unipr.it
  - Scegliere pool Ingegneria Lab, Informatica XY

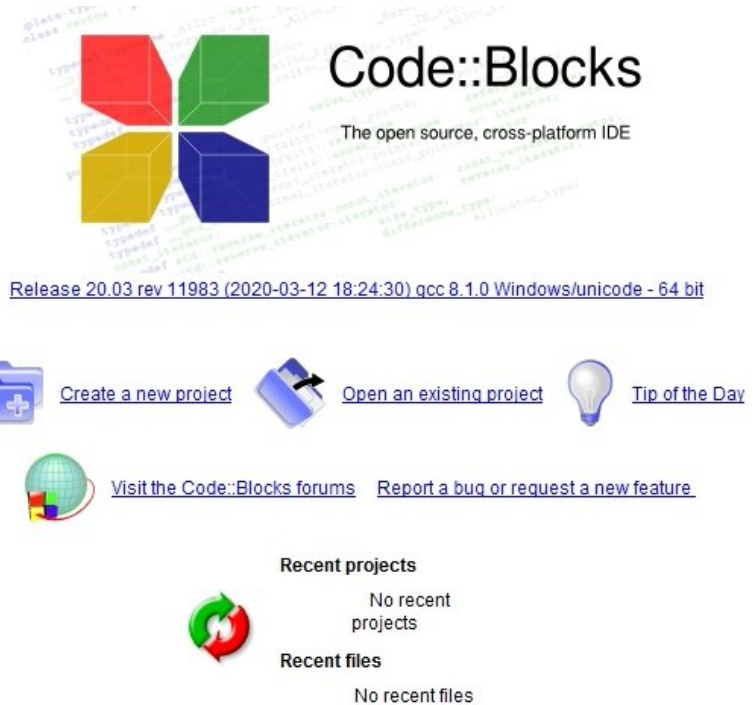
- Ambiente integrato
- IDE: Integrated Development Environment
  - Gestione progetti
  - Editor
  - Compilatore (e linker)
  - Debugger
- Release 20.03 (luglio 2022)



- Prima esecuzione:
  - Installazione
  - Scegliere default



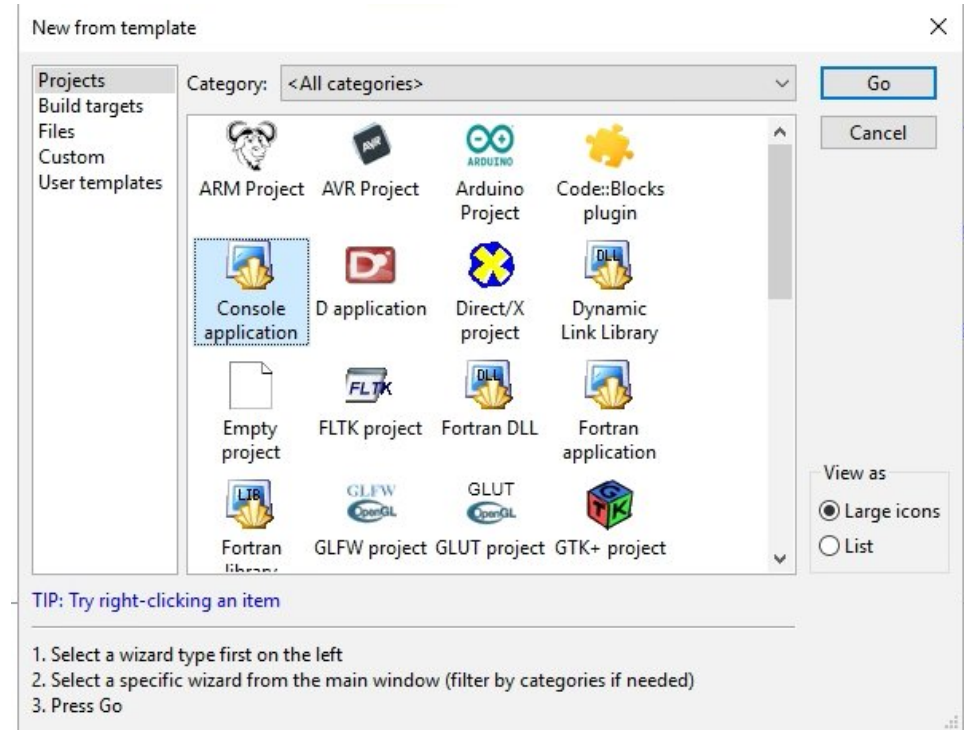
- Creare il vostro codice:
  - Create New Project



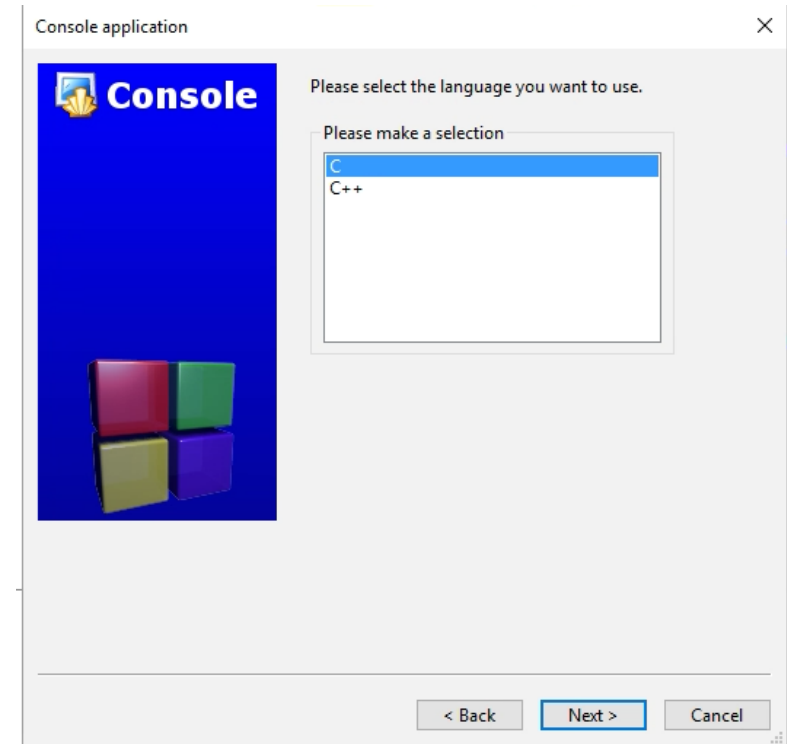


# Code::blocks: creazione nuovo progetto

- Creare il vostro codice:
  - Create New Project
  - Console application



- Creare il vostro codice:
  - Create New Project
  - Console application
    - Wizard
  - Select C

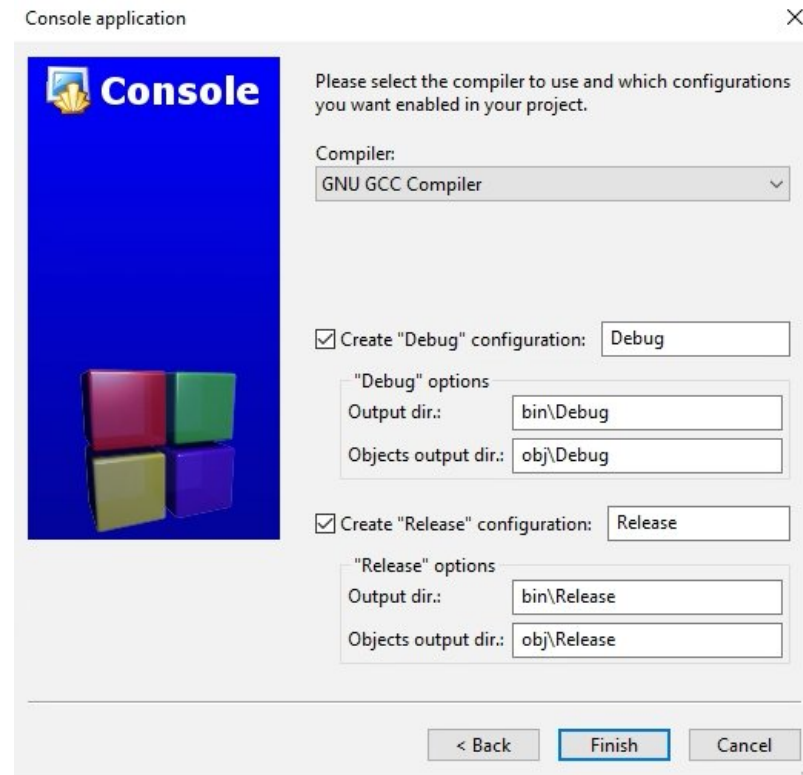


- Creare il vostro codice:
  - Create New Project
    - Wizard
  - Console application
  - Select C
  - Inserire
    - Titolo progetto
    - Folder (questo PC → U:)

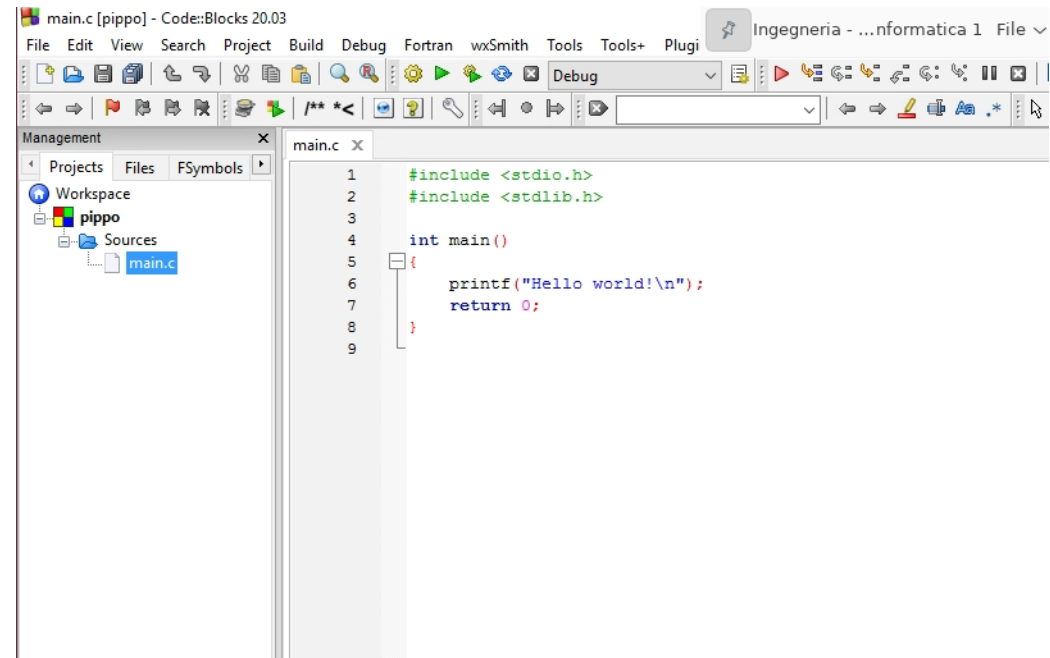


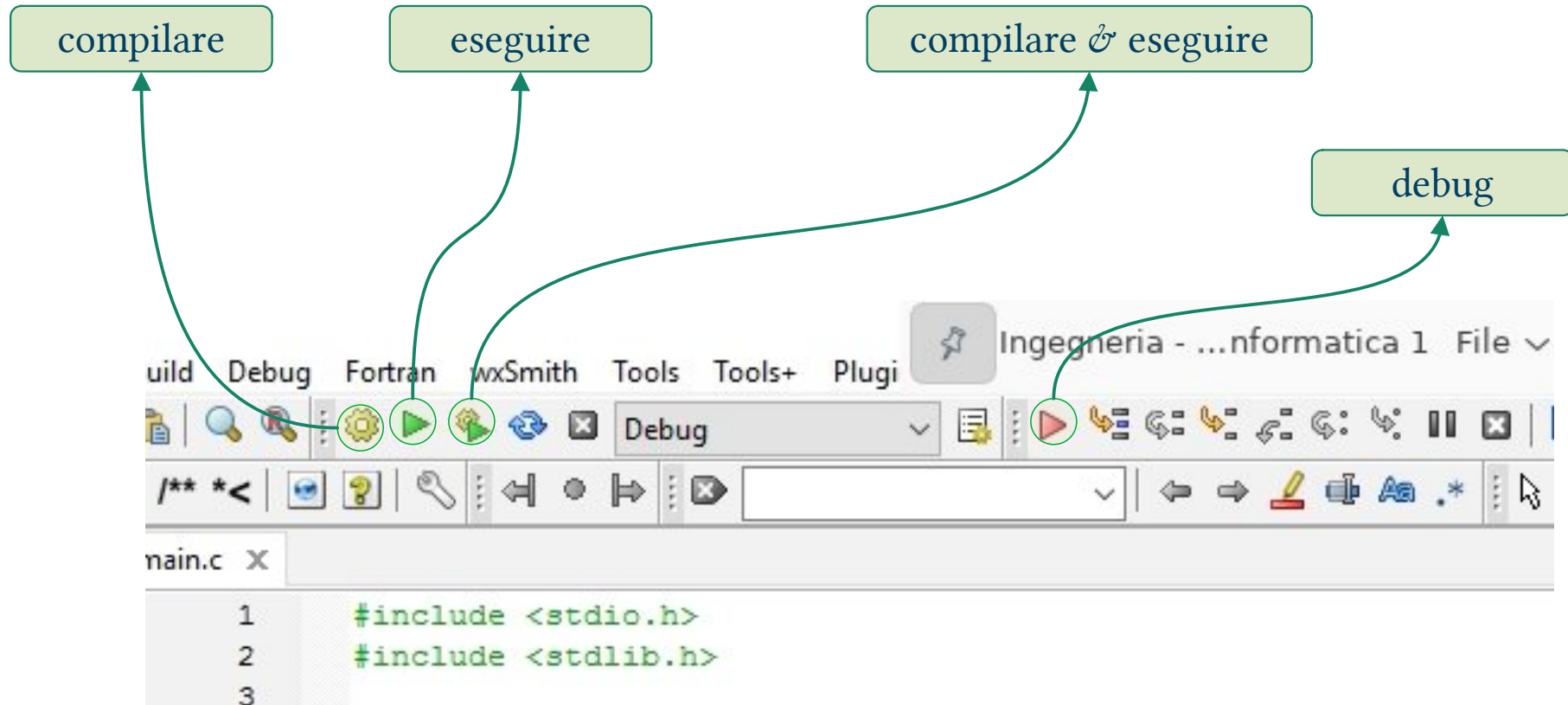
# Code::blocks: creazione nuovo progetto

- Creare il vostro codice:
  - Create New Project
    - Wizard
  - Console application
  - Select C
  - Inserire
    - Titolo progetto
    - Folder (questo PC → U:)
  - Scelta compilatore → default

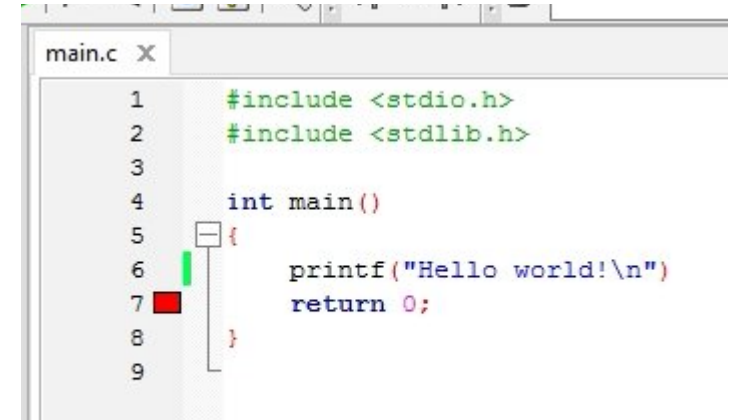


- Alla fine della procedura:
  - Progetto
  - Scheletro main.c

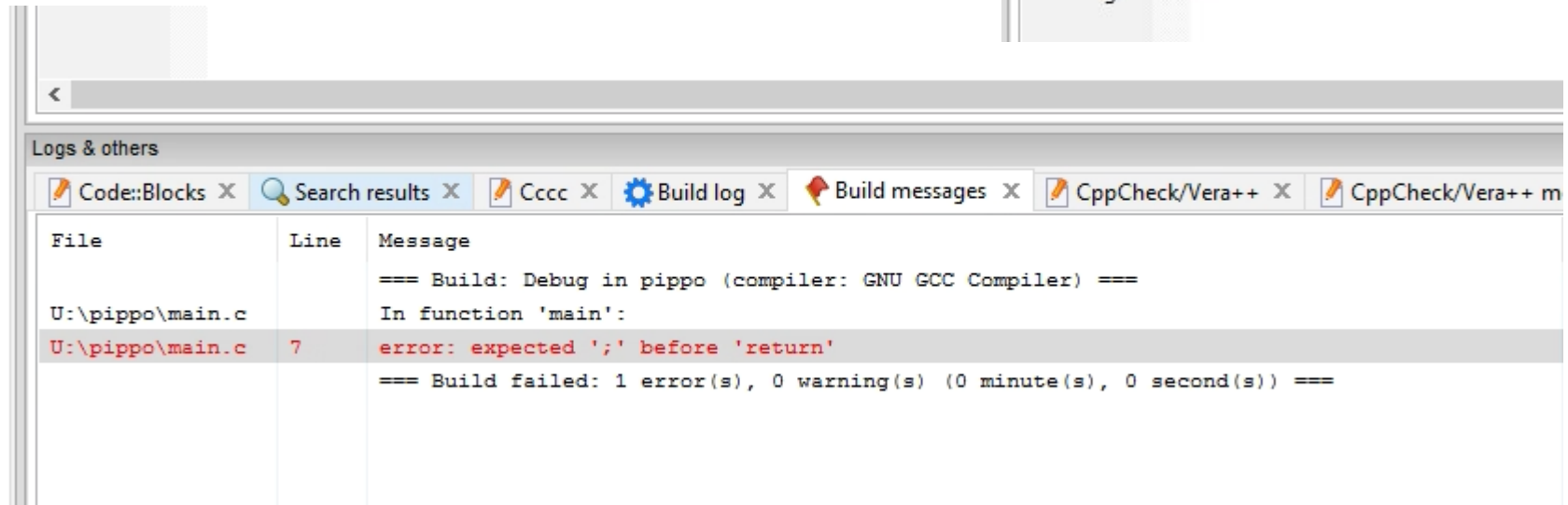




- Genera eseguibile
- Attenzione a warning ed errori!

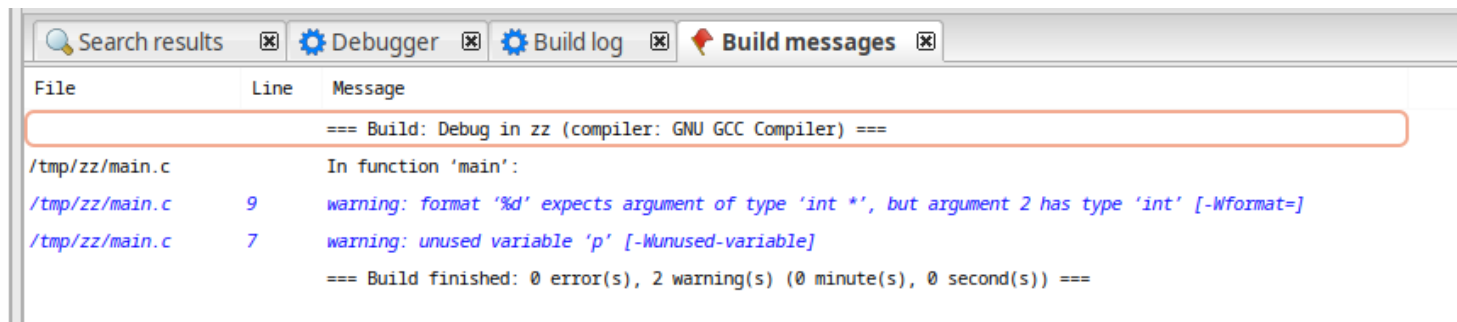


```
main.c X
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main()
5  {
6      printf("Hello world!\n");
7      return 0;
8  }
9
```



| Logs & others                                                                                           |      |                                     |
|---------------------------------------------------------------------------------------------------------|------|-------------------------------------|
| Code::Blocks X Search results X Cccc X Build log X Build messages X CppCheck/Vera++ X CppCheck/Vera++ m |      |                                     |
| File                                                                                                    | Line | Message                             |
| === Build: Debug in pippo (compiler: GNU GCC Compiler) ===                                              |      |                                     |
| In function 'main':                                                                                     |      |                                     |
| U:\pippo\main.c                                                                                         | 7    | error: expected ';' before 'return' |
| === Build failed: 1 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===                               |      |                                     |

- Errore di sintassi o lessicale → errore di compilazione
  - Non si scappa, dovete risolverlo!
- Qualcosa non suona bene ma è ammesso → compila ma warning
  - **Assolutamente NON ignorare, vanno risolti!**

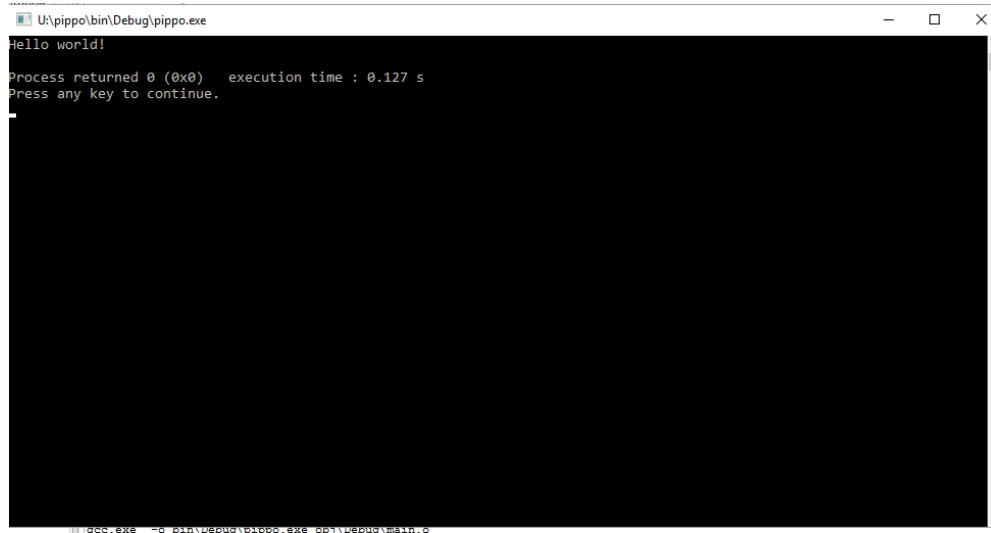


The screenshot shows the 'Build messages' window in Code::Blocks. It displays the output of a GCC compilation. The window has tabs for 'Search results', 'Debugger', 'Build log', and 'Build messages'. The 'Build messages' tab is active, showing a table with columns 'File', 'Line', and 'Message'. The messages indicate a successful build with two warnings: a format specifier warning on line 9 and an unused variable warning on line 7. The build summary at the bottom shows 0 errors and 2 warnings.

| File                                                                        | Line | Message                                                                                          |
|-----------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------|
| === Build: Debug in zz (compiler: GNU GCC Compiler) ===                     |      |                                                                                                  |
| /tmp/zz/main.c                                                              |      | In function 'main':                                                                              |
| /tmp/zz/main.c                                                              | 9    | warning: format '%d' expects argument of type 'int *', but argument 2 has type 'int' [-Wformat=] |
| /tmp/zz/main.c                                                              | 7    | warning: unused variable 'p' [-Wunused-variable]                                                 |
| === Build finished: 0 error(s), 2 warning(s) (0 minute(s), 0 second(s)) === |      |                                                                                                  |



- Esecuzione
  - Apre “console”
  - NON chiudere con mouse

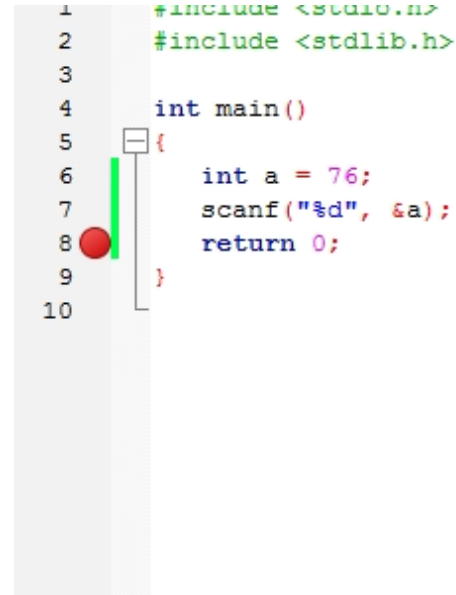


```
U:\pippo\bin\Debug\pippo.exe
Hello world!
Process returned 0 (0x0)   execution time : 0.127 s
Press any key to continue.
```

The screenshot shows a standard Windows console window. The title bar indicates the file path 'U:\pippo\bin\Debug\pippo.exe'. The console output displays 'Hello world!', followed by a status message 'Process returned 0 (0x0) execution time : 0.127 s' and a prompt 'Press any key to continue.'. The window is currently waiting for a key press to close.

- Modalità debug
  - Esecuzione passo passo
  - Ispezione valori variabili
  - Breakpoint
- ~~In laboratorio NON funziona~~

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main()
5  {
6      int a = 76;
7      scanf("%d", &a);
8      return 0;
9  }
10
```



| Watches |         |     |  |
|---------|---------|-----|--|
| a       | intext! | ... |  |
|         |         |     |  |

- Possibile a lezione
  - NON ad esame
- Installare la stessa versione di code::blocks
  - WINDOWS: con MinGW
  - LINUX: occorre gcc
  - MAC OS: ...



## Microsoft Windows

### File

codeblocks-20.03-setup.exe  
codeblocks-20.03-setup-nonadmin.exe  
codeblocks-20.03-nosetup.zip  
codeblocks-20.03mingw-setup.exe  
codeblocks-20.03mingw-nosetup.zip  
codeblocks-20.03-32bit-setup.exe  
codeblocks-20.03-32bit-setup-nonadmin.exe  
codeblocks-20.03-32bit-nosetup.zip  
codeblocks-20.03mingw-32bit-setup.exe  
codeblocks-20.03mingw-32bit-nosetup.zip

### Download from

[FossHUB](#) or [Sourceforge.net](#)  
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- Pro:
  - Più veloce (forse)
  - Pratico per esercitarsi a casa
- Contro:
  - All'esame non utilizzabile
  - Occorre impratichirsi comunque all'ambiente per prova d'esame



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# Laboratori & code::blocks



**KEEP  
CALM**  
IT'S  
**QUESTION  
TIME**

*Per me si va ne la città dolente,  
per me si va ne l'eterno dolore,  
per me si va tra la perduta gente*

Dante Alighieri, Inferno Canto III